



AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

**Dept. of Computer Science
Faculty of Science and Technology**

CSC2210: OBJECT ORIENTED PROGRAMMING 2

SPRING 2025-2026

Section: [K]

**Project Report On
[AIUB BUS MANAGEMENT SERVICE]**

**Supervised By
Dr.Md.Iftekharul Mobin**

Submitted By:

NAME	STUDENT ID
MD MORSALIN HASAN MIM	23-54497-3

Obtained Marks for CO2 and CO3 (Description given in the following page)					
Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria (CO2)	Total =		Evaluation Criteria (CO3)	Total =	
Requirement fulfillment			Organization of the application		
Validation			Representation and Integration of Database		
Verification			Graphical User Interface		

Introduction:

The **AIUB Bus Management System** is a comprehensive software solution designed to modernize and streamline the daily transportation operations of the university. Managing a large-scale bus fleet involves complex coordination between administrative oversight, student requirements, and driver logistics. This project addresses these challenges by replacing manual record-keeping with a secure, automated platform that ensures efficiency and transparency for all stakeholders. The system is built with three distinct user modules: an **Admin Panel** for high-level management of routes, students, and drivers; a **Student Panel** for route selection and fee payments; and a **Driver Panel** for managing schedules and confirming pickups. By integrating real-time data tracking—such as payment status and student boarding counts—the system minimizes operational errors and provides a reliable transit experience. Ultimately, this project aims to enhance the university's transit ecosystem through a user-friendly digital interface that saves time and improves service reliability.

AIUB Bus Management System - Database Design

Case-Study:

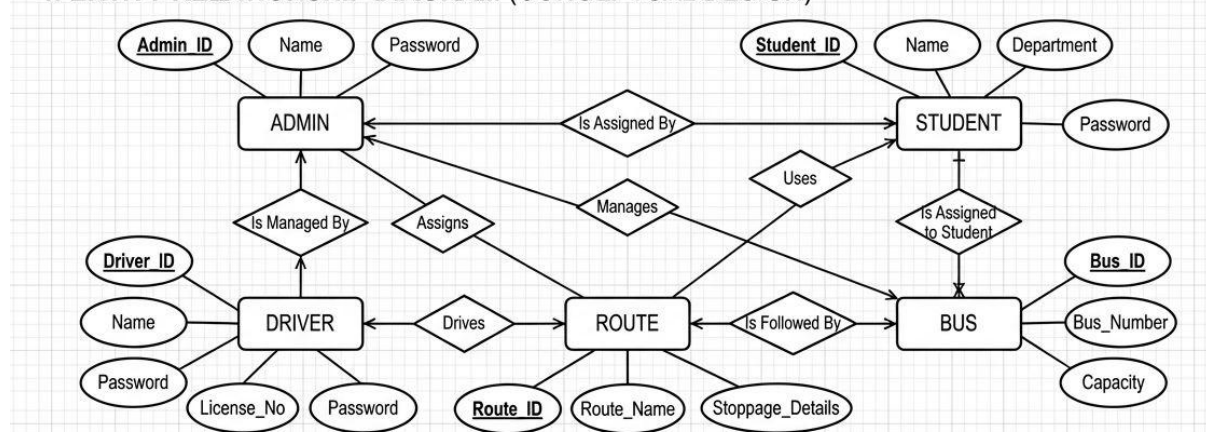
The AIUB Bus Management System is a digital solution designed to streamline university transportation by replacing manual coordination with a centralized database. The system operates through three primary portals—Admin, Student, and Driver—where the Admin manages all fleet assignments, and users access their specific details via secure login.

The database architecture relies on five key entities: Admin, Student, Driver, Bus, and Route. Relationships are managed using Foreign Keys, allowing the Admin to link drivers to buses and buses to specific routes. This relational design ensures high Data Integrity and Security, providing an efficient, automated experience for the university community.

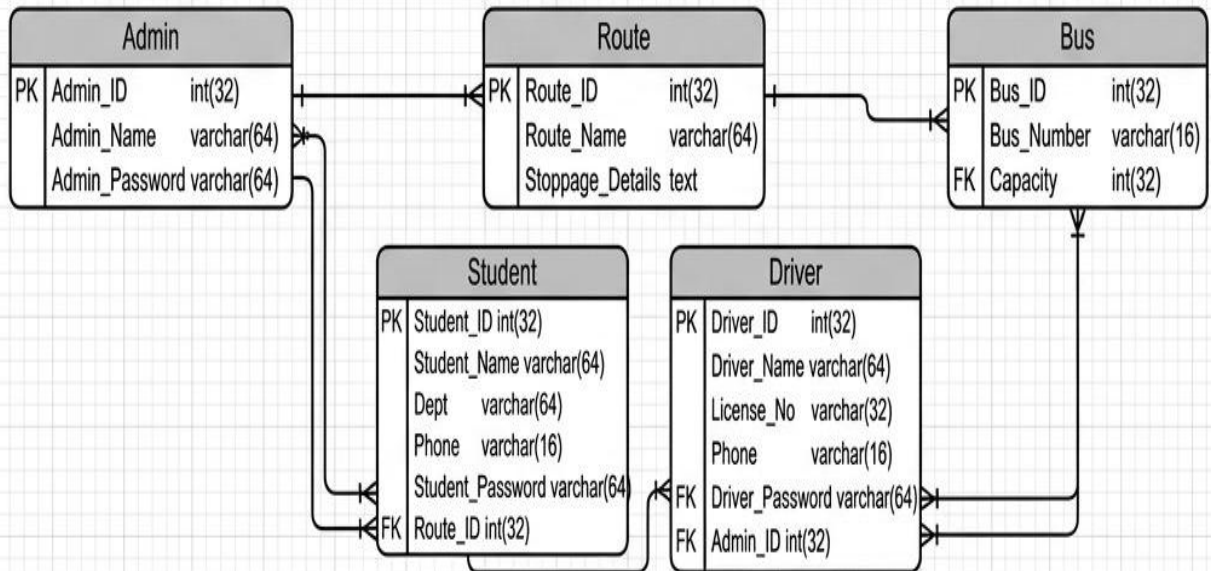
1. Entity-Relationship (ER) Diagram Description

- **Entities:** Admin, Student, Driver, Bus, Route.
- **Attributes:**
 - **Admin:** Admin_ID (PK), Admin_Name, Admin_Password.
 - **Student:** Student_ID (PK), Student_Name, Dept, Phone, Student_Password, Route_ID (FK).
 - **Driver:** Driver_ID (PK), Driver_Name, License_No, Phone, Driver_Password, Bus_ID (FK), Admin_ID (FK).
 - **Bus:** Bus_ID (PK), Bus_Number, Capacity, Route_ID (FK).
 - **Route:** Route_ID (PK), Route_Name, Stoppage_Details.
- **Relationships:**
 - Admin **Manages** Student, Driver, Bus, and Route.
 - Driver **Drives** a specific Bus.
 - Bus **Follows** a specific Route.
 - Student **Uses** a Route assigned by the Admin.

1. ENTITY-RELATIONSHIP DIAGRAM (CONCEPTUAL DESIGN)



2. DATABASE SCHEMA (RELATIONAL DESIGN)



2. Database Schema (Relational Tables)

:

Table: Admin		
Key Type	Field Name	Data Type
PK	Admin_ID	int(32)
	Admin_Name	varchar(64)
	Admin_Password	varchar(64)
Table: Route		
Key Type	Field Name	Data Type
PK	Route_ID	int(32)
	Route_Name	varchar(64)
	Stoppage_Details	text
Table: Student		
Key Type	Field Name	Data Type
PK	Student_ID	int(32)
	Student_Name	varchar(64)

Table: Student		
Key Type	Field Name	Data Type
	Dept	varchar(64)
	Phone	varchar(16)
	Student_Password	varchar(64)
FK	Route_ID	int(32)
Table: Driver		
Key Type	Field Name	Data Type
PK	Driver_ID	int(32)
	Driver_Name	varchar(64)
	License_No	varchar(32)
	Phone	varchar(16)
	Driver_Password	varchar(64)
FK	Bus_ID	int(32)
FK	Admin_ID	int(32)
Table: Bus		
Key Type	Field Name	Data Type
PK	Bus_ID	int(32)
	Bus_Number	varchar(16)
	Capacity	int(32)
FK	Route_ID	int(32)

SQL Query:

Create Database

```
CREATE DATABASE Transport_System;
USE Transport_System;
```

Admin Table

```
CREATE TABLE Admin (
  Admin_ID INT PRIMARY KEY,
  Admin_Name VARCHAR(64),
  Admin_Password VARCHAR(64));
```

Route Table

```
CREATE TABLE Route (
  Route_ID INT PRIMARY KEY,
  Route_Name VARCHAR(64),
  Stoppage_Details TEXT );
```

Bus Table

```
CREATE TABLE Bus (
  Bus_ID INT PRIMARY KEY,
  Bus_Number VARCHAR(16),
  Capacity INT,
```

```

Route_ID INT,
FOREIGN KEY (Route_ID) REFERENCES
Route(Route_ID)
);

```

Student Table

```

CREATE TABLE Student (
Student_ID INT PRIMARY KEY,
Student_Name VARCHAR(64),
Dept VARCHAR(64),
Phone VARCHAR(16),
Student_Password VARCHAR(64),
Route_ID INT,
FOREIGN KEY (Route_ID) REFERENCES
Route(Route_ID)
);

```

Driver Table

```

CREATE TABLE Driver (
Driver_ID INT PRIMARY KEY,
Driver_Name VARCHAR(64),
License_No VARCHAR(32),
Phone VARCHAR(16),
Driver_Password VARCHAR(64),
Bus_ID INT,
Admin_ID INT,
FOREIGN KEY (Bus_ID) REFERENCES
Bus(Bus_ID),
FOREIGN KEY (Admin_ID) REFERENCES
Admin(Admin_ID)
);

```

Insert Sample Data

```

INSERT INTO Admin VALUES (1, 'Admin1',
'1234');

```

```

INSERT INTO Route VALUES (1, 'Route A',
'Stop1, Stop2, Stop3');

```

```

INSERT INTO Bus VALUES (1, 'BUS-101', 40,
1);

```

```

INSERT INTO Student VALUES (1, 'Rahim',
'CSE', '01711111111', 'pass123', 1);

```

```

INSERT INTO Driver VALUES (1, 'Karim',
'LIC12345', '01822222222', 'driver123', 1, 1);

```

Show All Students

```

SELECT * FROM Student;

```

Show Students with Route Name

```

SELECT Student.Student_Name,
Route.Route_Name

```

```

FROM Student
JOIN Route ON Student.Route_ID =
Route.Route_ID;

```

Show Bus with Route

```

SELECT Bus.Bus_Number, Route.Route_Name
FROM Bus
JOIN Route ON Bus.Route_ID =
Route.Route_ID;

```

Show Driver with Bus and Admin

```

SELECT Driver.Driver_Name,
Bus.Bus_Number, Admin.Admin_Name
FROM Driver
JOIN Bus ON Driver.Bus_ID = Bus.Bus_ID
JOIN Admin ON Driver.Admin_ID =
Admin.Admin_ID;

```

Update Query

```

UPDATE Student
SET Phone = '01999999999'
WHERE Student_ID = 1;

```

Delete Query

```

DELETE FROM Student
WHERE Student_ID = 1;

```

Full System View

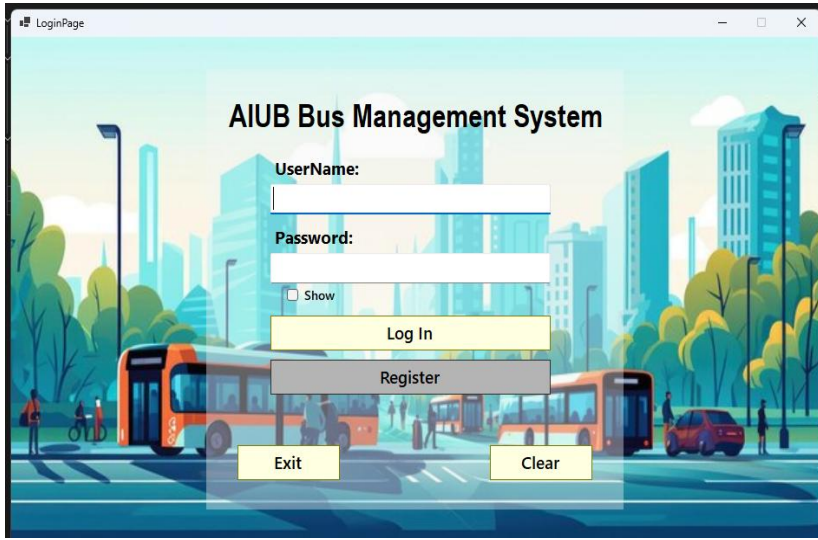
```

SELECT
Student.Student_Name,
Route.Route_Name,
Bus.Bus_Number,
Driver.Driver_Name
FROM Student
JOIN Route ON Student.Route_ID =
Route.Route_ID
JOIN Bus ON Route.Route_ID = Bus.Route_ID
JOIN Driver ON Bus.Bus_ID = Driver.Bus_ID;

```

Screenshot of forms:

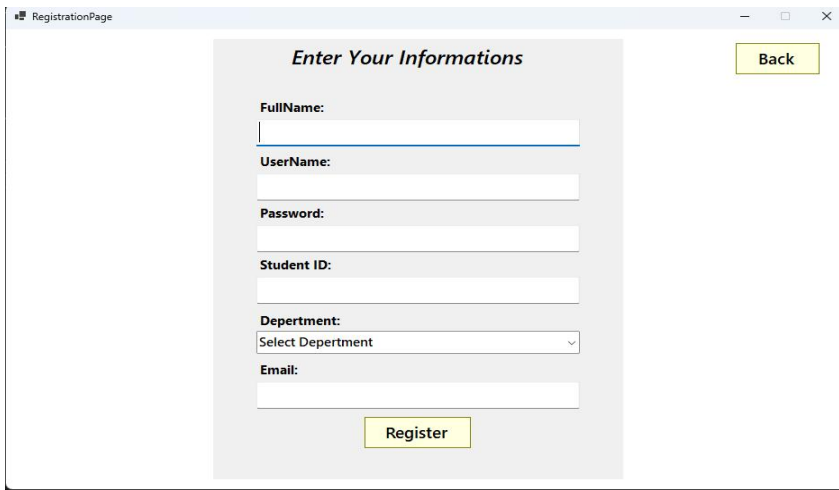
REGISTRATION AND USERLOGIN DASHBOARD



The screenshot shows a web application window titled "LoginPage". The main heading is "AIUB Bus Management System". Below the heading, there are two input fields: "UserName:" and "Password:". A checkbox labeled "Show" is positioned below the password field. There are four buttons: "Log In" (yellow), "Register" (grey), "Exit" (yellow), and "Clear" (yellow). The background is a colorful illustration of a city street with a bus, trees, and buildings.

Login Page

The Login Page serves as the initial entry point for all users of the system. It features a clean interface where users must provide their unique User-name and Password to gain access to their respective panels. The page includes a Show checkbox to help users verify their password entry, along with a Log In button to authenticate the session. For navigation and ease of use, there is a Register button for new users, a Clear button to reset the input fields, and an Exit button to close the application.



The screenshot shows a web application window titled "RegistrationPage". The main heading is "Enter Your Informations". Below the heading, there are six input fields: "FullName:", "UserName:", "Password:", "Student ID:", "Department:" (a dropdown menu with "Select Department" as the selected option), and "Email:". There are two buttons: "Register" (yellow) and "Back" (yellow). The background is a light grey gradient.

2)Registration Page : The **Registration Page** allows new students to join the system by providing their essential credentials. Users are required to fill out a form with their **FullName**, a chosen **UserName**, and a secure **Password**. To ensure academic verification, the form also includes fields for the **Student ID**, a dropdown menu to **Select Department**, and a primary **Email** address. Once all information is entered, the user can click the **Register** button to create their account. Additionally, a **Back** button is available to return to the main login screen if needed.

ADMIN DASHBORD

AIUB Bus Management System Admin

Manage Routes Manage Students Manage Driver Assign Bus Reports

Route Name: Select Direction: Create Route

Location Name: Add Location

Id	RouteName	Direction
4	Mirpur Route	ToUniversity
5	Uttara Route	ToUniversity
6	Dhanmondi Route	FromUniversity
7	Badda Route	ToUniversity
8	mohammadpur	ToUniversity

Id	RouteId	LocationName
7	4	Mirpur 10
8	4	Mirpur 12
9	4	Kalshi
14	4	Badda

1)The Route Management section allows the admin to define and organize the geographical network of the bus service. Admins can Create Routes by naming them—such as the Mirpur, Uttara, or Dhanmondi routes—and assigning a Direction (either ToUniversity or FromUniversity). Furthermore, the system supports Location Mapping, where specific stops like Mirpur 10, Kalshi, or Badda can be added to a specific RouteID to ensure accurate student pickups.

AIUB Bus Management System Admin

Manage Routes Manage Students Manage Driver Assign Bus Reports

Total Collected Amount: 30.00 Refresh

Id	Date	StudentName	RouteName	TimeSlot	Amount	IsPaid	PickupConfirmed
4	5/12/2026	Morsalin	mohammadpur	09:00 - 10:00	10.00	☐	☐
3	1/20/2026	Salman Student	Badda Route	14:00 - 16:00	10.00	☐	☐
2	1/20/2026	Nabila Student	Uttara Route	10:00 - 12:00	10.00	☐	☐
1	1/20/2026	Salman Student	Mirpur Route	08:00 - 10:00	10.00	☒	☒

2) The Main Dashboard serves as the central hub for monitoring real-time system activity. It prominently displays the Total Collected Amount, which currently shows a sum of 30.00. Below this summary, a comprehensive transaction table tracks student bookings, providing details such as the Booking ID, Date, Student Name, Route, and Time Slot. This allows the admin to monitor financial flow and verify if a student's payment (IsPaid) and Pickup have been officially confirmed

AIUB Bus Management System Admin

Manage Routes Manage Students Manage Driver Assign Bus Reports

Mirpur Route Karim Driver 08:00 - 10:00 5/12/2026 Assign

Date	RouteName	DriverName	TimeSlot
2026-05-12	mohammadpur	mhm12	08:00 - 10:00

3)The final operational stage involves the Assign Bus and Reports features. The Assign Bus function allows the admin to link a specific Driver to a Route and a designated Time Slot, such as the 08:00-10:00 window. **Finally, the Reports section is available for the admin to review overall system performance, focusing on trip history and total financial collections to ensure the system is running efficiently**

AIUB Bus Management System Admin

Manage Routes Manage Students Manage Driver Assign Bus Reports

Search Students Search Active Block

Id	Name	Username	StudentId	Department	Email	Status
12	Salman Student	salman	12-12125-1	CSE	salman@student.aiub...	Active
13	Nabila Student	nabila	20-12346-1	BBA	nabila@student.aiub...	Active
14	Morsalin	mhm10	123	CSE	mhm1013@gmail...	Active

4) In the Manage Students module, the admin has full authority over user access. The interface features a Student Directory that lists all registered users with their Username, Full Name, Student ID, Department (such as CSE or BBA), and Email. To maintain security and order, the admin can use the Search function to find specific records and toggle a student's account status between Active and Block to manage their eligibility for the service.

AIUB Bus Management System Admin

Manage Routes Manage Students Manage Driver Assign Bus Reports

Full Name: Username: Password: Add Driver

Licence No.: Mobile No.: Remove

Id	Name	Username	LicenceNo	Phone	Status
10	Rahim Driver	rahim	DL-1001	01711111111	Active
11	Karim Driver	karim	DL-1002	01722222222	Active
16	mhm12	mhm12	101	00000	Active

5)The Manage Driver section is dedicated to maintaining a qualified team of operators. Through this interface, the admin can perform Driver Registration by inputting the driver's Full Name, Username, Password, License Number, and Mobile Number. The Driver Database keeps a live record of all staff, such as Rahim Driver and Karim Driver, showing their Phone details and Status. If a driver is no longer part of the system, the admin can use the Remove button to delete their profile.

Student Dashboard

The screenshot shows the 'Pickup Selection' module of the AIUB Bus Management System. It features a 'Time Slot' dropdown set to '08:00 - 10:00' and a 'Routes' dropdown set to 'mohammadpur'. Below these, there is a 'Pickup Locations' section with four radio button options: 'mohammadpur bus stand' (selected), 'bosila', 'asadgate', and 'bijoysononi'. A 'Save Pickup' button is located at the bottom right of the form.

1. Pickup Selection

The **Pickup Selection** module allows students to organize their daily commute by selecting their preferred travel parameters. Through the interface, a student can choose a specific **Time Slot** (such as **08:00 - 10:00**) and a designated **Route**, like **mohammadpur**. Once the route is selected, the system provides a list of available **Pickup Locations**, including options like **mohammadpur bus stand**, **bosila**, **asadgate**, and **bijoysononi**. After making their selection, the student can finalize their request by clicking the **Save Pickup** button.

The screenshot shows the 'Profile Update' module of the AIUB Bus Management System. It displays the student's current information: **Student ID: 123**, **Department: CSE**, and **Email: mhmim13@gmail**. There is an 'Update Profile' button at the bottom.

3. Profile Update

The **Profile Update** module is dedicated to personal information management within the system. In this section, students can view and modify their essential academic and contact details, such as their **Student ID** (e.g., **123**), **Department** (e.g., **CSE**), and registered **Email** address. Once any necessary changes have been made to these fields, the student can click the **Update Profile** button to save the new information and keep their account data current.

The screenshot shows the 'Fee Payment' module of the AIUB Bus Management System. It displays the 'Total Due: 0.00' and a 'Pay Amount' input field. A 'Pay' button is located to the right of the input field. Below this, there is a table for tracking payment history.

Date	Amount	Status
------	--------	--------

2. Fee Payment

The **Fee Payment** section provides a streamlined way for students to manage their transport finances. This interface displays the **Total Due** amount and allows the student to enter a specific **Pay Amount** before clicking the **Pay** button to process the transaction. Additionally, a transaction table is included to track payment history, showing the **Date**, **Amount**, and **Status** of each payment to ensure clear financial records for the student.

Driver Dashboard

AIUB Bus Management System Driver Logout

View Assigned Routes Confirm Pickup's

Today's Assigned Buses Refresh

RouteName	Direction	Time	Date
mohammadpur	ToUniversity	08:00 - 10:00	5/10/2026

Id	RouteId	LocationName
17	8	mohammadpur...
18	8	bosila
19	8	asadgate
20	8	bijoysononi

1. View Assigned Routes : The **View Assigned Routes** section allows drivers to access their daily schedules and operational details. The interface features a table titled **Today's Assigned Buses**, which lists critical information such as the **Route Name** (e.g., mohammadpur), travel **Direction** (ToUniversity), assigned **Time** (08:00 - 10:00), and the specific **Date**. Additionally, a secondary table displays the sequence of stops for the route, including the **Location Name** and associated **Route ID**, such as mohammadpur, bosila, asadgate, and bijoysononi. Drivers can use the **Refresh** button to update this list and ensure they have the most current assignment data.

AIUB Bus Management System Driver Logout

View Assigned Routes Confirm Pickup's

Time Slot: 08:00 - 10:00 Confirm

LocationName	StudentCount
asadgate	0
bijoysononi	0
bosila	1
mohammadpur bus stand	0

2. Confirm Pickups

The **Confirm Pickup's** module is designed to help drivers manage student boarding along their designated route. By selecting a specific **Time Slot**, the driver can view a list of pickup locations and the corresponding **Student Count** for each stop. For example, the system might indicate that there is **1 student** waiting at **bosila** while other locations have **0**. Once the driver arrives at a location and completes the boarding process, they can use the **Confirm** button to update the system, providing real-time tracking for the administration.

Some SQL Database

100 % ✓ No issues found

Results Messages

	Id	Name	Username	Password	Role	Status
1	9	Admin	admin	1234	Admin	Active
2	10	Rahim Driver	rahim	1234	Driver	Active
3	11	Karim Driver	karim	1234	Driver	Active
4	12	Salman Student	salman	1234	Student	Active
5	13	Nabila Student	nabila	1234	Student	Active
6	14	Morsalin	mhm10	123456	Student	Active
7	16	mhm12	mhm12	1234	Driver	Active

Results Messages

	UserId	StudentId	Department	Email
1	12	12-12125-1	CSE	salman@student.aiub.edu
2	13	20-12346-1	BBA	nabila@student.aiub.edu
3	14	123	CSE	mhmim13@gmail.com

100 % ✓ No issues found

Results Messages

	Id	RouteId	DriverId	TimeSlotId	Date
1	4	4	10	1	2026-01-20
2	5	5	11	2	2026-01-20
3	6	6	10	3	2026-01-20
4	7	7	11	4	2026-01-20
5	8	8	16	1	2026-05-12

Results Messages

	Id	StartTime	EndTime
1	1	08:00:00.0000000	10:00:00.0000000
2	2	10:00:00.0000000	12:00:00.0000000
3	3	12:00:00.0000000	14:00:00.0000000
4	4	14:00:00.0000000	16:00:00.0000000
5	5	16:00:00.0000000	18:00:00.0000000

Results Messages

	Id	RouteName	Direction
1	4	Mirpur Route	ToUniversity
2	5	Uttara Route	ToUniversity
3	6	Dhanmondi Route	FromUniversity
4	7	Badda Route	ToUniversity
5	8	mohammadpur	ToUniversity

Results Messages

	Id	StudentId	LocationId	TimeSlotId
1	1	12	7	1
2	2	12	8	1
3	3	13	10	2
4	4	13	11	2
5	5	12	13	3
6	6	12	16	4
7	7	14	18	1

Features and Facilities

- **Centralized Administration:** The system provides a powerful admin dashboard to manage the entire bus network, including monitoring total collections and managing student access.
- **Dynamic Route Planning:** Admins can easily create new routes, set directions such as "ToUniversity" or "FromUniversity," and map specific pickup locations like Mirpur 10 or Badda.
- **Student Self-Service:** Students have the flexibility to select their preferred time slots, routes, and specific pickup points directly from their own dashboard.
- **Streamlined Fee Payment:** The platform includes a dedicated payment module where students can view their total due amount and process payments online.
- **Driver Schedule Management:** Drivers receive real-time updates on their daily assigned buses, routes, and time schedules to ensure punctual operations.
- **Boarding Confirmation:** To ensure accuracy, drivers can view the student count for each stop and confirm pickups, which automatically updates the admin's records.
- **Profile and Security Management:** The system ensures secure access through a login/registration process and allows users to keep their academic and contact information up to date

Conclusion

In conclusion, the **AIUB Bus Management System** effectively bridges the gap between university administration, students, and transport personnel, facilitating seamless daily commutes and enhancing overall organizational efficiency. By providing a user-friendly platform for route selection, automated fee payment, and real-time driver coordination, the system streamlines the entire transportation process, ensuring a reliable service for the university community. The integrated tracking of payments and confirmed pickups highlights the success of this collaborative digital model. Overall, the efficient operation of this system contributes to a highly organized and hassle-free experience for everyone involved in the university's transit ecosystem.

END